

Braided Notions of Dialogue Categories: reworked proofs

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1 Dialogue Categories: Basic Constructions and Facts

Definition 1.1. Let $\mathcal{C}$ be a monoidal category, a $\mathcal{C}$ is *dialogue category* if there is an exponentiable object $\perp$, i.e., there are bijections natural in A and B

$$\begin{aligned}\varphi_{A,B} : \mathcal{C}(A \otimes B, \perp) &\cong (B, A \multimap \perp) \\ \psi_{A,B} : \mathcal{C}(A \otimes B, \perp) &\cong (A, \perp \multimap B)\end{aligned}$$

Here are some useful combinators:

- The left evaluation map:

$$\text{leval}_A = \varphi_{A, A \multimap \perp}^{-1} 1_{A \multimap \perp} : A \otimes (A \multimap \perp) \rightarrow \perp$$

- The right evaluation map:

$$\text{reval}_A = \psi_{\perp \multimap A, A}^{-1} 1_{\perp \multimap A} : (\perp \multimap A) \otimes A \rightarrow \perp$$

- The composite left evaluation map:

$$\begin{aligned}\text{lev}_{B,A} : B \otimes ((A \otimes B) \multimap \perp) &\rightarrow (A \multimap \perp) \\ \text{lev}_{B,A} &= \varphi_{A, B \otimes ((A \otimes B) \multimap \perp)} (\text{leval}_{A \otimes B} \circ \alpha_{A, B, A \otimes B \multimap \perp})\end{aligned}$$

- The composite right evaluation map:

$$\begin{aligned}\text{rev}_{A,B} : (\perp \multimap (A \otimes B)) \otimes A &\rightarrow (\perp \multimap B) \\ \text{rev}_{A,B} &= \psi_{(A \otimes B \multimap \perp) \otimes A, B} (\text{reval}_{A \otimes B} \circ \alpha_{A \otimes B \multimap \perp, A, B})\end{aligned}$$

Definition 1.2. Let $\mathcal{C}$ be a dialogue category,

- a *turn* in $\mathcal{C}$ is a natural isomorphism $\text{turn}_A : A \multimap \perp \cong \perp \multimap A$.

*The original text is by Paul-André Mellies

- a *wheel* in $\mathcal{C}$ is a bijection $\mathcal{C}(A \otimes B, \perp) \simeq \mathcal{C}(B \otimes A, \perp)$ natural in both A and B .

Theorem 1.1. There is a 1-to-1 correspondence between turns and wheels in any dialogue category $\mathcal{C}$.

Proof. Given a turn $\text{turn}_A : A \multimap \perp \cong \perp \multimap A$ natural in A , define $\text{wheel}_{A,B} : \mathcal{C}(A \otimes B, \perp) \rightarrow \mathcal{C}(B \otimes A, \perp)$ as the following composite of bijections:

$$\mathcal{C}(A \otimes B, \perp) \xrightarrow{\varphi_{A,B}} \mathcal{C}(B, A \multimap \perp) \xrightarrow{\mathcal{C}(1_B, \text{turn}_A)} \mathcal{C}(B, \perp \multimap A) \xrightarrow{\psi_{A,B}^{-1}} \mathcal{C}(B \otimes A, \perp)$$

This is obviously invertible, just take $\varphi_{A,B}^{-1} \circ \mathcal{C}(B, A \multimap \perp) \circ \psi_{A,B}$. *wheel* is also natural in both arguments. Take $f : A_1 \rightarrow A_2$ and $g : A_2 \otimes B \rightarrow \perp$, then the following square commutes:

$$\begin{array}{ccc} B \otimes A_1 & \xrightarrow{1_B \otimes f} & B \otimes A_2 \\ & \searrow \text{wheel}(g \circ f \otimes 1_B) & \swarrow \text{wheel}_{A_2, B}(f) \\ & \perp & \end{array}$$

First of all, we have the following by the naturality of internal homs:

$$\text{turn}_{A_1} \circ \mathcal{C}(f, \perp) = \mathcal{C}(f, \perp) \circ \text{turn}_{A_2} \quad (1)$$

Further, observe that:

$$\psi_{B, A_1}^{-1}(\text{turn}_{A_1} \circ \varphi_{A_1, B}(g \circ f \otimes 1_B)) = \psi_{B, A_2}^{-1}(\text{turn}_{A_2} \circ \varphi_{A_2, B}(g)) \circ (1_B \otimes f) \quad (2)$$

Since

$$\begin{aligned} \psi_{B, A_1}^{-1}(\text{turn}_{A_1} \circ \varphi_{A_1, B}(g \circ f \otimes 1_B)) &= \psi_{B, A_1}^{-1}(\text{turn}_{A_1} \circ \mathcal{C}(f, 1_\perp) \circ \varphi_{A_2, B}(g)) \\ &= \psi_{B, A_1}^{-1}(\mathcal{C}(f, 1_\perp) \circ \text{turn}_{A_2} \circ \varphi_{A_2, B}(g)) = \psi_{B, A_2}^{-1}(\text{turn}_{A_2} \circ \varphi_{A_2, B}(g)) \circ (1_B \otimes f) \end{aligned}$$

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